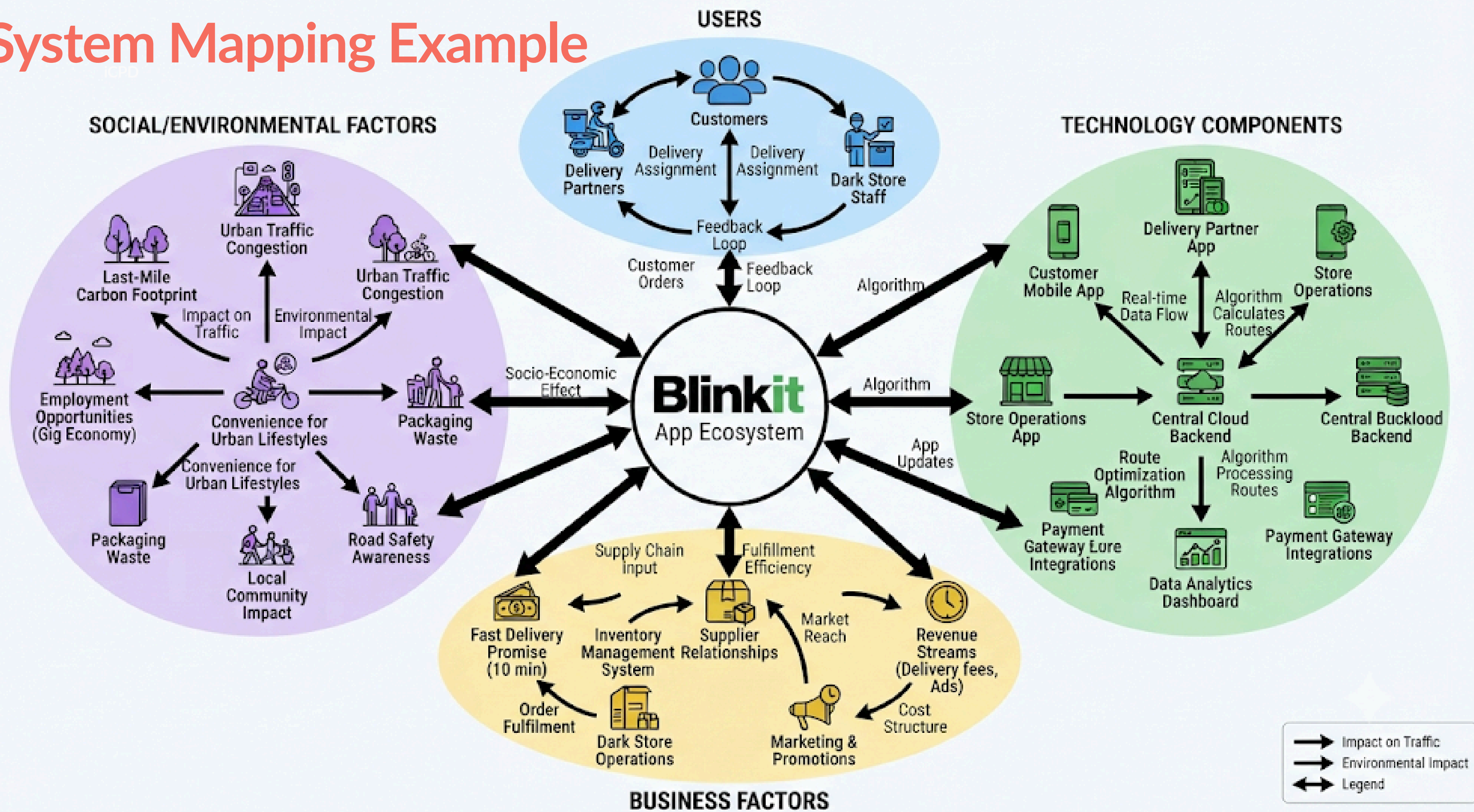


LECTURE 7

System Mapping Example



LECTURE 8

Food Delivery App Discounts

Reinforcing Loop (R)

More discounts

- More users order
- More data collected
- Better recommendations
- More orders

Growth happens quickly.

Balancing Loop (B)

Too many discounts

- Company loses revenue
- Reduces discount budget
- Orders slow down

This loop limits growth.

Lesson: Every successful product has both loops simultaneously.

Traffic Congestion

Reinforcing Loop (R)

More cars on road

- More congestion
- Slower travel
- People leave earlier
- Even more cars at same time

Congestion worsens.

Balancing Loop (B)

More congestion

- Government builds metro/public transport
- Fewer cars
- Congestion reduces

System tries to stabilize.

Lesson: Every successful product has both loops simultaneously.

Exam Pressure

Reinforcing Loop (R)

High competition

- More coaching classes
- Higher expectations
- More pressure
- Even more competition

Stress escalates.

Balancing Loop (B)

High stress

- Students seek breaks/counseling
- Stress reduces
- Performance stabilizes

Lesson: Every successful product has both loops simultaneously.

Why Causal Loops Matter in Product Design

Key Insight to Remember

- **Reinforcing Loops = Growth or Collapse**
- **Balancing Loops = Stability or Resistance**
- **Delays = Surprises**

Most real systems are a mix of both, which is why predicting outcomes is hard.

Activity 2: “System Mapping Sprint”

15 Minutes

Prompts

- What happens if delivery time increases?
- Who gets affected first?
- What data flows exist?

What happens if delivery time increases?

Impact on Customer Behavior & Retention

- **Reduced Order Frequency:** Studies on quick-commerce patterns show that when a delivery arrives late or takes longer, consumers subconsciously wait longer before placing their next order.
- **Shifting Expectations:** While users still expect fast service, a departure from strict ultra-fast windows forces platforms to prioritize consistency and reliability over a literal countdown timer to prevent frustration.

What happens if delivery time increases?

Shift in Brand Strategy & Marketing

- **Pivoting Away from Speed Guarantees:** Platforms often drop explicit time-bound promises (such as the traditional "10-minute" badge) to protect themselves against public backlash, regulatory scrutiny, and supply chain bottlenecks.
- **Focus on Scale and Range:** Instead of marketing sheer velocity, marketing messages naturally pivot toward broader inventory selection, product availability, and overall convenience at the doorstep.

What happens if delivery time increases?

Operational & Delivery Partner Relief

- **Decreased Pressure on Workers:** Relaxing strict time limits directly mitigates safety hazards, lowering the risk of rash driving and reducing physical stress on gig-workers.
- **Regulatory Compliance:** Slower or more flexible delivery windows align more closely with government and labor ministry concerns regarding responsible time-bound corporate promises.

Activity 2: “System Mapping Sprint”

15 Minutes

Prompts

- What happens if delivery time increases?
- Who gets affected first?
- What data flows exist?

Who Gets Affected First?

- Delivery Partners (Gig Workers)
- The Customers
- Dark Store Operations Staff

Who Gets Affected First?

Delivery Partners (Gig Workers):

- **Direct Impact:** They experience the friction on the ground immediately—whether they are stuck in traffic, facing bad weather, or waiting longer at the dark store for an order to be packed.
- **Operational Stress:** Slower deliveries disrupt their batching schedules, reducing their hourly earnings potential and increasing physical fatigue.

Who Gets Affected First?

The Customers:

- **Immediate Awareness:** Almost simultaneously, the customer notices the shift through the app interface as the Estimated Time of Arrival (ETA) ticks upward or misses the standard ultra-fast window.
- **Behavioral Shift:** This sparks immediate frustration, leading to a spike in customer support chats or a drop in trust.

Who Gets Affected First?

Dark Store Operations Staff:

- **Internal Pressure:** If the delay originates from a store bottleneck (such as slow packing or inventory mismatches), the floor staff face immediate pressure from the management dashboard to clear the queue.

Activity 2: “System Mapping Sprint”

15 Minutes

Prompts

- What happens if delivery time increases?
- Who gets affected first?
- What data flows exist?

Customer ----->> Central Backend:

Transmits order placement data, cart contents, payment status, and user geo-location permissions.

Central Backend ----->> Dark Store System:

Instantly pushes order details, triggers automated inventory deductions, and generates a prioritized item-picking list for store staff.

Dark Store Staff ----->> Delivery Partner App:

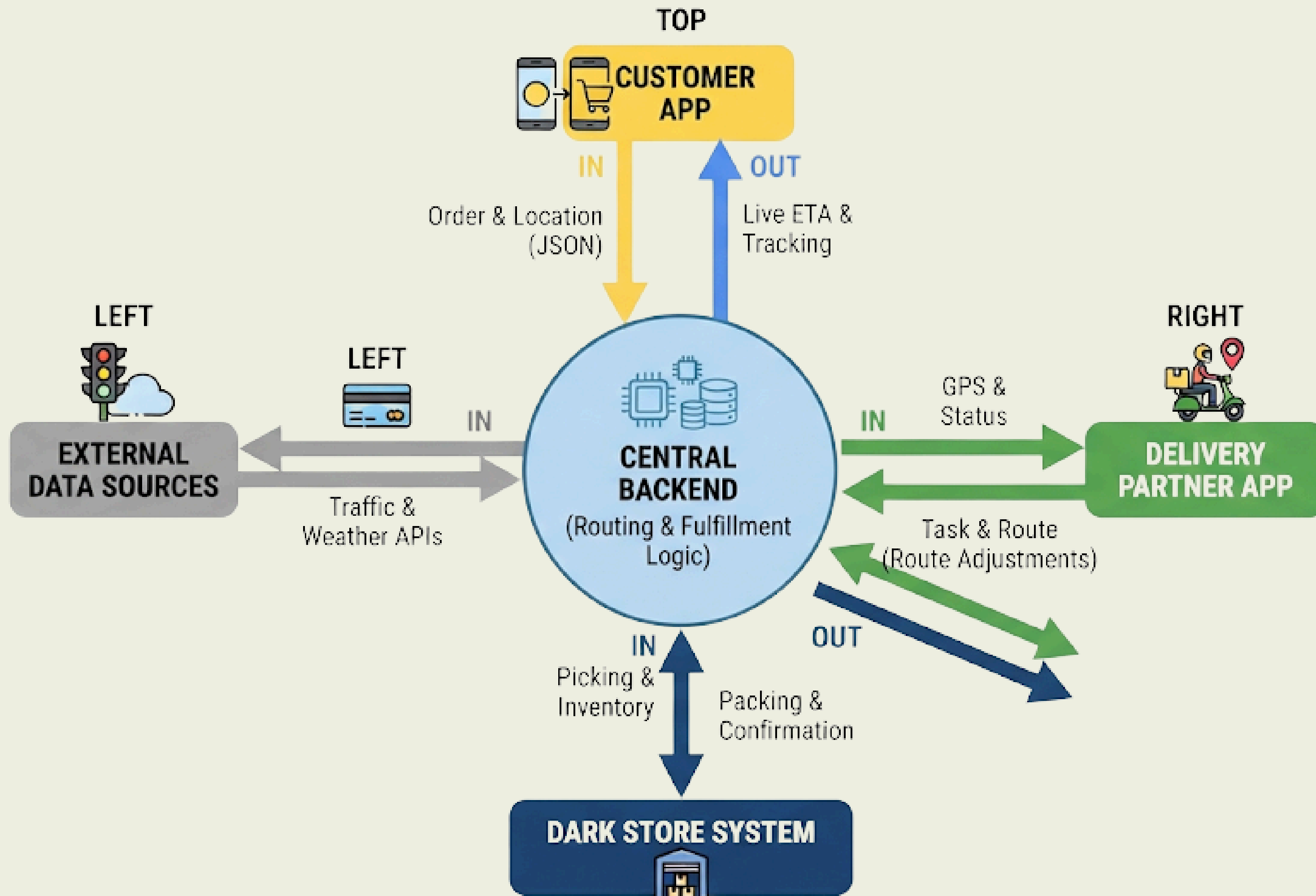
Signals when an order is packed and hands off the delivery task notification to the nearest available rider.

Delivery Partner App ----->> Central Backend (Real-Time Telemetry):

Continuously transmits GPS coordinates, speed, and route progress (via WebSockets) so the system knows where the rider is at all times.

Central Backend ----->> Customer App (Dynamic ETAs):

If traffic or delays occur, routing algorithms recalculate the path and push updated time estimates back to the user's tracking screen.



Why Causal Loops Matter in Product Design

Engineers often think:

“If we add Feature X, users will be happy.”

Systems Thinking asks:

- What happens after that?
- What happens because of that?
- What happens later?

Without Causal Loops

You design features.

With Causal Loops

You design behaviors and outcomes.

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The “Black Box” Concept

In engineering and system design, we often treat a system as a "Black Box." This means we focus on:

Inputs: What goes into the system (data, energy, raw materials).

Process: What the system does to the inputs (transformation).

Outputs: What comes out (information, heat, finished products).

Emergence: The “Magic” of System

One of the most important concepts in system theory is *Emergence*. This is the idea that the system possesses properties that its individual parts do not have.

An individual neuron cannot "think," but a system of billions of neurons creates "consciousness." A single molecule of water isn't "wet," but a system of them is.

System Design

Imagine you open a small lemonade stand. It's easy. You have one pitcher and one cup. Suddenly, a celebrity tweets about your lemonade. Now, 10,000 people are standing in your driveway.

- *You can't just work faster. You'll collapse.*
- *You can't just get a bigger pitcher. Eventually, they don't make pitchers big enough.*
- *System Design is the plan to hire 100 people, set up 20 stands, and create a system so they all have the same lemonade and no one waits too long.*

System Design

When you "design" a system, you are essentially arranging these "Lego bricks" in a specific pattern:

Load Balancer (The Traffic Cop): When people arrive, a "Traffic Cop" stands at the front and points them to different lines. "You go to Stand 1, you go to Stand 2." This ensures no single stand gets overwhelmed.

Caching (The "Pre-made" Pile): Instead of squeezing a lemon every time someone asks, you keep 50 cups already poured and sitting on ice. It's much faster to hand someone a pre-made cup than to make it from scratch.

Database Sharding (The Filing Cabinets): If you have a list of a billion customers, it's too heavy for one book. You split it up: "Customers A-M are in this book; N-Z are in that book." Now, two people can look up names at the same time.